Transcription

10/30/26

Progeria - accelerated aging disorder

- Caused by a single base change

Codons are only in translation.

The enzyme moves along the double helix catalyzing the reactions that synthesize an RNA polymer. Specifically, NTPs enter a channel in the enzyme that leads to its active site. If one of the NTPs matches the next base in the template DNA, which is also in the active site, the enzyme catalyzes the formation of a phosphodiester bond.

RNA polymerase makes an RNA version of the information encoded in a gene after initial proteins open the promoter and unwind the double helix.

- specialized to catalyze the formation of phosphodiester bonds that create a new RNA molecule and complete transcription

Transcription is…

- An enzyme-catalyzed reaction that makes phosphodiester linkages between ribonucleotides that are complementary to the bases in a DNA template
- A polymerization reaction
- Spontaneous because the substrates involved (ribonucleotide triphosphates) have high potential energy

Since DNA is double-stranded, the strand that is read by RNA polymerase and is transcribed is called the template strand, and the complementary strand that is not transcribed is called the non-template (coding) strand.

Promoter - a region of the gene where RNA polymerase makes initial contact with DNA to start transcription; also determines the direction in which the strand is read

The part of the gene that RNA polymerase has already transcribed is called upstream, and the part it hasn't yet gotten to is called downstream.

